

PHY 517 / AST 443: Observational Techniques in Astronomy

Lecture:

Proposals
Time Allocation Committee

Telescope time / data analysis proposals

- writing (successful) proposals is an essential part of being a researcher
- ... at the latest, when you need to apply for funding
- observational astronomers need to submit proposals for telescope time
- can also submit proposals for funding to analyze existing data

Example: Hubble Space Telescope

- proposal deadline once per year (~April)
- typically ~1000 proposals: time, archival, theory
- ~20% success rate
- open to anyone

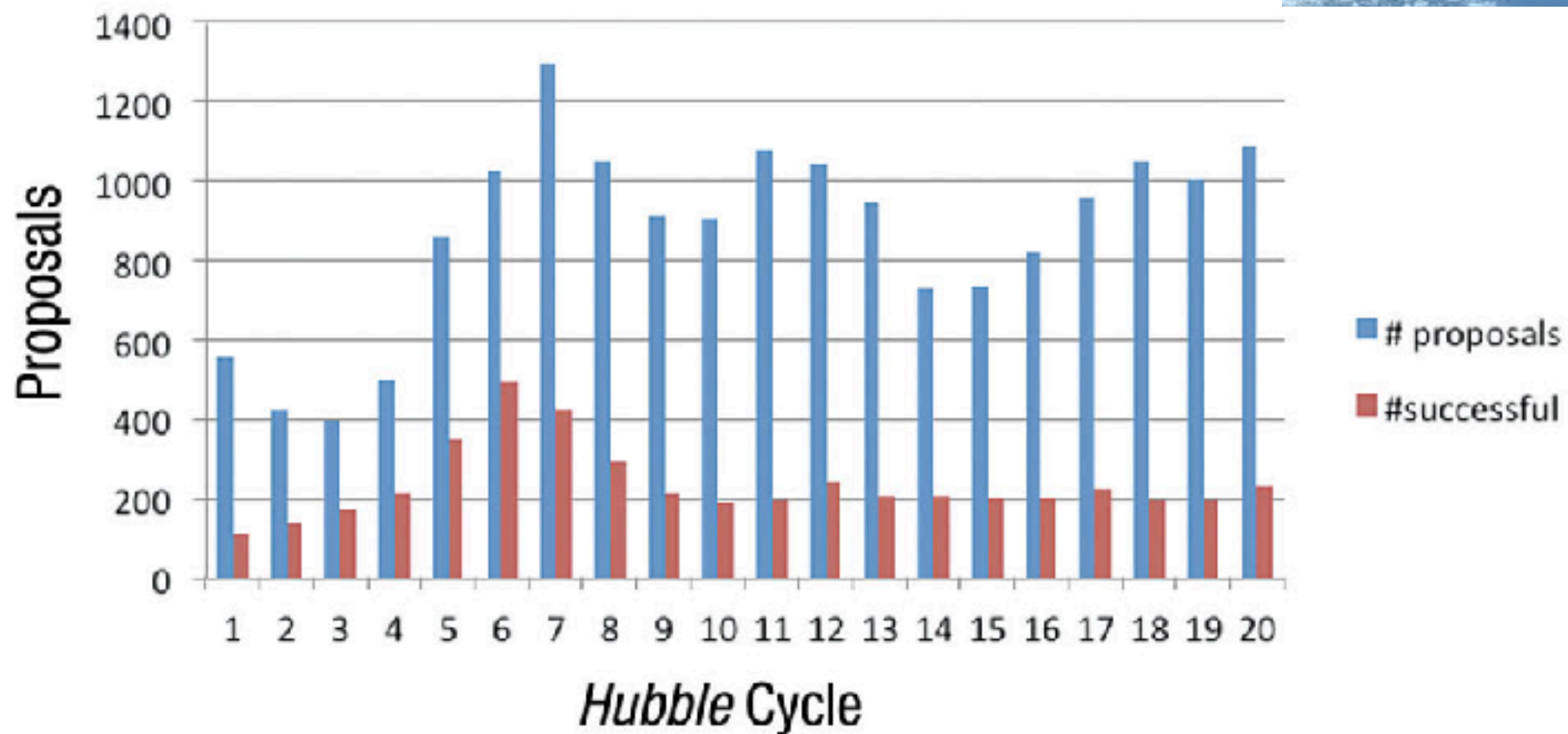
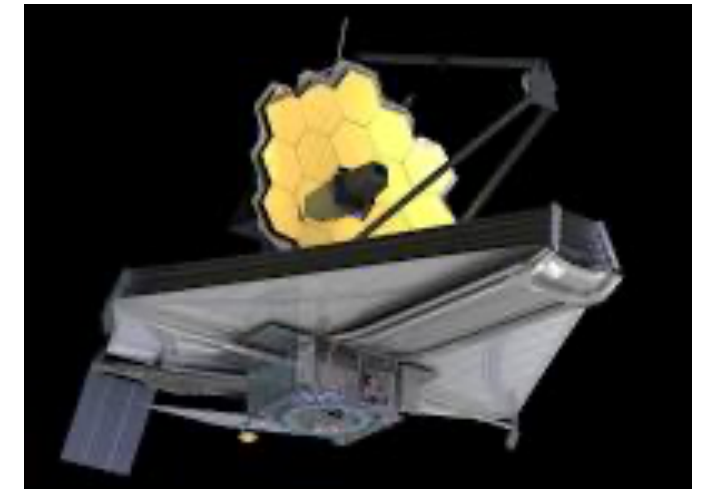


Figure 1: *Hubble* proposal pressure by number of proposals. The blue histogram shows the number of proposals submitted each cycle; the red shows the number accepted. The Cycle 7 statistics include the cycle 7N and 7AR proposals. The oversubscription ranges from 2:1 in Cycle 6 to more than 5:1 since SM4.

JWST



- proposal deadline once per year
- last Cycle 5 (Oct 15th 2025)
- Cycle 4: 2377 proposals: time, archival, theory
- ~10% success rate
- open to anyone
- proposals ~1x per year

ID ▼	Program Title ▼	PI & Co-PIs ▼	Exclusive Access Period (months) ▼	Prime/Parallel Time (hours) ▼	Instrument/Mode ▼	Type ▼
2875	Scrutinizing the Dirtiest Cepheids, a Test of the Hubble Tension	PI: Adam Riess	12	15.97/1.26	NIRCam/Imaging	GO
2974	The Hubble constant at 1.9% from spatially resolved kinematics of gravitational lens	PI: Tommaso Treu Co-PI: Anwar J. Shajib	12	36.3/0	NIRSpec/IFU	GO
3055	A TRGB Calibration of Surface Brightness Fluctuations	PI: R. Tully	12	46.83/22.06	NIRCam/Imaging	GO
3325	Mapping the Most Extreme Protoclusters in the Epoch of Reionization	PI: Feige Wang Co-PI: Jinyi Yang	12	44.51/16.34	NIRCam/WFSS NIRSpec/IFU NIRSpec/MOS	GO
3950	Unlocking the Early Universe for Weak Lensing with JWST: High-Precision Analysis of z=2 Galaxy Cluster XLSSC122	PI: Kyle Finner	12	4.99/0	NIRCam/Imaging	GO
4265	Unveiling the interplay between the circumgalactic and interstellar media in a complex protocluster environment at z=4.5	PI: Jorge Gonzalez Lopez Co-PI: Manuel Aravena	12	17.03/0	NIRCam/Imaging NIRSpec/IFU NIRSpec/MOS	GO

link to proposal guidelines: https://jwst-docs.stsci.edu/jwst-opportunities-and-policies/jwst-call-for-proposals-for-cycle-3#JWSTCallforProposalsforCycle3-TOC_anchor

NOIRLab

- National Optical-Infrared Astronomy Research Laboratory
- formerly NOAO = National Optical Astronomical Observatories
- US national research & development center for ground-based night-time astronomy
- manages most telescopes with US-wide access
- own facilities: Kitt Peak National Observatory (KPNO, Arizona), Cerro-Tololo Inter-American Observatory (CTIO, Chile)
- Calls for Proposals 2x per year; deadlines end of September and end of March



ESO

- ESO = European Southern Observatory
- manages the Very Large Telescope (VLT; Chile)
- also open to anyone; preference for European projects only in direct conflicts
- Calls for Proposals 2x per year; deadlines end of September and end of March



ALMA

- ALMA = Atacama Large Millimeter Array
- multi-national project
- proposals through respective managing facilities, e.g. for US: NRAO = National Radio Astronomy Observatory
- proposals ~ 1 x per year



Other telescopes

- some facilities are not open-access, but only available to researchers at the institutions / countries who built / finance them
- for example:
 - Keck telescopes (mostly CalTech, University of California + University of Hawaii)
 - Subaru telescope (mostly Japan + University of Hawaii)



PHY517 / AST443 proposals

- each of you will write a *telescope proposal* for your Lab 3
- the telescope proposal has to be of the opposite observational technique (photometry/spectroscopy)
- you will then review the other students' proposals (~8 proposals to read, grade and evaluate)
- we will hold a Time Allocation Committee (TAC) meeting to discuss and rank the proposals
- each group will conduct their top-ranked project

PHY517 / AST443 proposals

- Proposal deadline: **Wed, October 14, 3:00pm (strict !)**
- Initial reviews deadline: **Friday, October 16, 11:59pm**
- Time Allocation Committee: **Monday, October 19**

How to write a good proposal

- come up with a good idea!
 - make sure that there is a *measurement* involved (i.e. not just plotting a lightcurve)
- figure out the technical details

Possible resource: AAVSO

American Association of Variable Star Observers (AAVSO):

- “alert”: call for observations by small telescopes, issued by scientists
- could pick one (or more) of these as basis for your proposal
- try to maximize science output (1 single 10-minute observations probably not very exciting)
- note: you will have to research the topic to write your science case

AAVSO.org/aauso-alert-notice-for-observing-campaigns-and-discoveries

AAVSO Alert Notices for Observing Campaigns and Discoveries

Note: This page, together with the AAVSO Target Tool Alerts/Campaigns target list, replaces the following AAVSO webpages: AAVSO Alert Notice Archive, AAVSO Special Notice Archive, and the original Observing Campaigns webpage. - July 2017

[Click here for Active Alert Notices.](#)

There are two types of AAVSO Alert Notices:

- **Observing Campaign**
 - to announce an **observing campaign** of short or long duration on one or more astronomical objects, at the request of an astronomer or the AAVSO;
 - to provide additional information about a campaign or the target of a previous Alert Notice
- **Object of Interest**
 - to announce the **discovery** of an object such as a nova or a bright supernova;
 - to report on noteworthy or unusual stellar behavior

An AAVSO Alert Notice for an **observing campaign** is issued as soon as a professional astronomer tells the AAVSO about a new, urgent need for special variable star observations, usually in connection with their research.

An AAVSO Alert Notice for an **object of interest** is issued in response to a discovery or other stellar activity that AAVSO HQ believes warrants in-depth coverage.

By subscribing* (free of charge) to the Alert Notices, you will receive them in your email inbox so that you can contribute crucial observations when they are most needed!

* To subscribe to the Alert Notices, log in to your AAVSO account (free), click "My account" at the top of the page, then click the "Profile" tab, then the "Email Settings" tab. Check the box next to "Alert Notices" and save your settings.

AAVSO Alert Notices are also posted on the AAVSO website.

Below are links to AAVSO Alert Notices, ordered by issue number with the most recent issue first. Links are also given to AAVSO Special Notices, which were issued when needed (now discontinued) to provide supplemental information about an observing campaign or a discovery, or to provide information about other stellar activity. Observers should use the page below, along with information in the AAVSO Target Tool Alerts/Campaigns target list (button at right), to see what targets are in need of observations to support current observing campaigns and to help plan their observing schedules.

Note that a Special Notice is located under the Alert Notice with which it is associated. Special Notices that are not associated with events covered in Alert Notices are listed at the end of the year in which they were issued. Please note that, as of July 2017, AAVSO Special Notices are no longer being issued.

The format below is date (yyyy/mm/dd), Alert or Special Notice number, Alert or Special Notice title/subject. (Missing dates to be added.)

For Alert Notices and associated Special Notices, jump to: Active, 2021, 2020, 2019, 2018, 2017, 2016, 2015, 2014, 2013, 2012, 2011, 2010, 2009, 2008, 2007, 2006, 2005, 2004, 2003, 2002, 2001, 2000, 1999, 1998, 1997, 1996, 1995, 1994, 1993, 1992, 1991, PEP Alert Notices, un-numbered Alert Notices

For AAVSO Special Notices that are not associated with AAVSO Alert Notices, jump to: SpN2017, SpN2016, SpN2015, SpN2014, SpN2013, SpN2012, SpN2011, SpN2010, SpN2009, SpN2008, SpN2007, SpN2006, SpN2005.

ACTIVE before the date means that the observing campaign is active or that the target of an object of interest Alert Notice warrants continued coverage. All active Alert Notices for campaigns or objects of interest are grouped together below. When the campaign is concluded or the target no longer warrants coverage, the Alert Notice will be moved to the year in which it was issued.

The text in square brackets following the title of the Alert Notice refers to the AAVSO Observing Section under which the target falls.

Active Alert Notices for Observing Campaigns and Objects of Interest

- **ACTIVE** 20210814 *Alert Notice 754* - Monitoring requested for 15 VY Scl asterodynamic variables in support of HST observations [CV, Spectroscopy]
- **ACTIVE** 20210823 *Alert Notice 753* - Monitoring requested for seven intermediate polars [CV]
- **ACTIVE** 20210809 *Alert Notice 752* - Rare Outburst of Recurrent Nova RS Ophiuchi [CV]
- **ACTIVE** 20210803 *Alert Notice 751* - RU Lup and BP Tau to be observed by HST and XMM-Newton [YSO]
- **ACTIVE** 20210803 *Alert Notice 750* - T CrB photometry and spectroscopy requested for HST and XMM-Newton observations [CV]
- **ACTIVE** 20210730 *Alert Notice 749* - Nova in Vulpecula: N Vul 2021 = TCP J20210770+2014003 [V606 Vul] [CV]
- **ACTIVE** 20210720 *Alert Notice 747* - V627 Psc photometry and spectroscopy requested [CV, Spectroscopy]
- **ACTIVE** 20210618 *Alert Notice 745* - High-resolution spectroscopy and photometry requested for rho Cas study [SPP, Spectroscopy]
- **ACTIVE** 20210814 *Alert Notice 745* - Nova in Hercules: N Her 2021 = TCP J18573085+1853386 = ZTF1 Baas1sjq [V1674 Her] [CV]
- **ACTIVE** 20210808 *Alert Notice 744* - V1117 Her observations requested [YSO]

Possible resource: AAVSO

American Association of Variable Star Observers (AAVSO):

- “alert”: call for observations by small telescopes, issued by scientists
- See Rubin alert stream
- could pick one (or more) of these as basis for your proposal
- try to maximize science output (1 single 10-minute observations probably not very exciting)
- note: you will have to research the topic to write your science case

aaavso.org/aaavso-alert-notice-for-observing-campaigns-and-discoveries

AAVSO Alert Notices for Observing Campaigns and Discoveries

Note: This page, together with the AAVSO Target Tool Alerts/Campaigns target list, replaces the following AAVSO webpages: AAVSO Alert Notice Archive, AAVSO Special Notice Archive, and the original Observing Campaigns webpage. - July 2017

[Click here for Active Alert Notices.](#)

There are two types of AAVSO Alert Notices:

- **Observing Campaign**
 - to announce an **observing campaign** of short or long duration on one or more astronomical objects, at the request of an astronomer or the AAVSO;
 - to provide additional information about a campaign or the target of a previous Alert Notice
- **Object of Interest**
 - to announce the **discovery** of an object such as a nova or a bright supernova;
 - to report on noteworthy or unusual stellar behavior

An AAVSO Alert Notice for an **observing campaign** is issued as soon as a professional astronomer tells the AAVSO about a new, urgent need for special variable star observations, usually in connection with their research.

An AAVSO Alert Notice for an **object of interest** is issued in response to a discovery or other stellar activity that AAVSO HQ believes warrants in-depth coverage.

By subscribing* (free of charge) to the Alert Notices, you will receive them in your email inbox so that you can contribute crucial observations when they are most needed!

* To subscribe to the Alert Notices, log in to your AAVSO account (free), click "My account" at the top of the page, then click the "Profile" tab, then the "Email Settings" tab. Check the box next to "Alert Notices" and save your settings.

AAVSO Alert Notices are also posted on the AAVSO website.

Below are links to AAVSO Alert Notices, ordered by issue number with the most recent issue first. Links are also given to AAVSO Special Notices, which were issued when needed (now discontinued) to provide supplemental information about an observing campaign or a discovery, or to provide information about other stellar activity. Observers should use the page below, along with information in the AAVSO Target Tool Alerts/Campaigns target list (button at right), to see what targets are in need of observations to support current observing campaigns and to help plan their observing schedules.

Note that a Special Notice is located under the Alert Notice with which it is associated. Special Notices that are not associated with events covered in Alert Notices are listed at the end of the year in which they were issued. Please note that, as of July 2017, AAVSO Special Notices are no longer being issued.

The format below is date (yyyy-mm-dd), Alert or Special Notice number, Alert or Special Notice title/subject. (Missing dates to be added.)

For Alert Notices and associated Special Notices, jump to: [Active](#), 2021, 2020, 2019, 2018, 2017, 2016, 2015, 2014, 2013, 2012, 2011, 2010, 2009, 2008, 2007, 2006, 2005, 2004, 2003, 2002, 2001, 2000, 1999, 1998, 1997, 1996, 1995, 1994, 1993, 1992, 1991, PEP Alert Notices, un-numbered Alert Notices

For AAVSO Special Notices that are not associated with AAVSO Alert Notices, jump to: [SpN2017](#), [SpN2016](#), [SpN2015](#), [SpN2014](#), [SpN2013](#), [SpN2012](#), [SpN2011](#), [SpN2010](#), [SpN2009](#), [SpN2008](#), [SpN2007](#), [SpN2006](#), [SpN2005](#).

ACTIVE before the date means that the observing campaign is active or that the target of an object of interest Alert Notice warrants continued coverage. All active Alert Notices for campaigns or objects of interest are grouped together below. When the campaign is concluded or the target no longer warrants coverage, the Alert Notice will be moved to the year in which it was issued.

The text in square brackets following the title of the Alert Notice refers to the AAVSO Observing Section under which the target falls.

Active Alert Notices for Observing Campaigns and Objects of Interest

- **ACTIVE** 20210814 *Alert Notice 754* - Monitoring requested for 15 VY Scl asterodynamic variables in support of HST observations [CV, Spectroscopy]
- **ACTIVE** 20210823 *Alert Notice 753* - Monitoring requested for seven intermediate polars [CV]
- **ACTIVE** 20210809 *Alert Notice 752* - Rare Outburst of Recurrent Nova RS Ophiuchi [CV]
- **ACTIVE** 20210803 *Alert Notice 751* - RU Lup and BP Tau to be observed by HST and XMM-Newton [YSO]
- **ACTIVE** 20210803 *Alert Notice 750* - T CrB photometry and spectroscopy requested for HST and XMM-Newton observations [CV]
- **ACTIVE** 20210730 *Alert Notice 749* - Nova in Vulpecular: N Vul 2021 = TCP J20210770+2014003 [V606 Vul] [CV]
- **ACTIVE** 20210720 *Alert Notice 747* - V627 Psc photometry and spectroscopy requested [CV, Spectroscopy]
- **ACTIVE** 20210618 *Alert Notice 745* - High-resolution spectroscopy and photometry requested for rho Cas study [SPP, Spectroscopy]
- **ACTIVE** 20210814 *Alert Notice 745* - Nova in Hercules: N Her 2021 = TCP J18573085+1853386 = ZTF1 Baas1ajq [V1674 Her] [CV]
- **ACTIVE** 20210808 *Alert Notice 744* - V1117 Her observations requested [YSO]

More resources on transient events

- Zwicky Transient Facility: <https://www.ztf.caltech.edu/>
- Latest supernovae: <https://www.rochesterastronomy.org/supernova.html>
- Transient Name Server: <https://www.wis-tns.org/>
- Rubin alert stream: <https://rubinobservatory.org/for-scientists/data-products/alerts-and-brokers>

Your data + archival data

- your observing proposal will be to make measurements with our roof-top telescope
- for the scientific analysis, you can *add* data from other sources
- examples:
 - AAVSO database -> lightcurve
 - spectroscopic information (Simbad, NED)
 - Gaia parallaxes -> distances
- see wiki tab “Astronomical Data Archives” for a list of resources

Possible project resource: basic astronomy

- think back to your introductory astronomy class (e.g. AST203)
- there are some “classic” measurements, e.g. color-magnitude diagrams, period-luminosity relations, etc.
- you could choose one of these
- make sure that there is a measurement goal!

Possible project resource: astro-news

- check astronomy news (e.g. Sky & Telescope)
 - also, what is currently visible (planets, comets, ...)
 - is there something that inspires you?
 - research the scientific background
-
- make sure that there is a measurement goal!
 - example: “*I want to make a pretty picture of this galaxy*” is not a quantitative measurement goal

Deep (stacked) imaging

- in the exoplanet lab, we analyzed each exposure separately
- can also combine exposures to create a deeper image with higher signal-to-noise
- straightforward with `scamp` command (provided you have run [astrometry.net](#) on your images) - see wiki page

Spectroscopy ?

- limited to bright targets that are easy to find or high surface brightness nebulae

Possible resource: Astronomical Ring for Access to Spectroscopy (ARAS):

- observing campaigns and alerts, specific for spectroscopy
- could pick one of these as basis for your proposal
- note: you will have to research the topic to write your science case
- read the spectroscopic manual of Lab 2!



Astronomical Ring for Access to Spectroscopy

An initiative dedicated to promotion of amateur astronomical spectroscopy and pro/am collaborations

[More info about ARAS...](#)



Last update: January 22, 2015

Programs and Surveys

[ARAS spectral data base](#)
Calculation of spectral lines,
novae, comets, ...

[HeSS data base](#)
The database of the stars catalogue
(GEM - Observatoire de Paris-
Meudon).

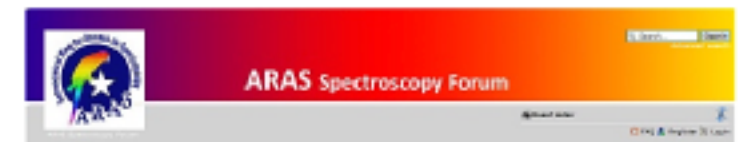
[ArasHeAm](#)
A front website to the data base for
help to select program stars.

[Be stars actuality](#)

[Novae spectroscopy
survey](#)

[FS CMa](#)

[Z CMa](#)



[Observations, technical infos, campaigns, beginners corner, and more....!](#)

[ARAS spectral data base](#)

[Comet C/2014 02 Lovejoy archive](#)

[T Tauri 2014-2015 campaign archive](#)

Possible project resource: *your own idea!*

- come up with your own idea!

Time Request

- the target observing period is October 20 - November 9
- you might need one full night, or several partial nights, or ...
- nights can be shared between groups
- try to be flexible in your time request
- I will assign observing time, based on proposal ranking and your (non-)availability
- examples of how to specify your time request:
 - 1 nights
 - 3x 2hours
- specify the nights you are *not* available (e.g. night before a mid-term / GRE)

Time Request

- if targeting transient sources: we cannot accommodate real Target-of-Opportunity requests (where your observations override somebody else's on a timescale of hours) - make sure you can do something with the nights you're given

Proposal Structure

- Cover Sheet
 - title
 - abstract
 - ~~PI and Col names~~
 - telescope / instrument / set-up request
 - time request
- Scientific Justification
 - limited to 1 page
- Figures, Tables, References (2 pages)
- Technical Justification / Experimental Design
 - limited to 1 page
- Analysis Plan
- Observing Run Details

Proposal Structure

telescopes such as ours.

Your data might even contribute to a journal paper! However, make sure that there is a measurement aspect to your project (i.e. not just plotting a lightcurve).

You can choose to add archival data (e.g. lightcurve data from AAVSO) to your own data to facilitate this.

- Other resources for transient events include [ZTF](#), the [Transient Name Server](#), and [this list of recent supernovae](#).
- For ideas specifically for spectroscopy, check [this forum](#).
- Think back of the concepts taught in your basic astronomy course. There are "standard" labs such as constructing color-magnitude diagrams. Make sure you have a quantitative science goal!
- Imaging analyses often acquire deep exposures of targets. It is straightforward to combine a set of exposures with the `{tt scamp}` program.
- For the data analysis, you can combine your data with external data. A list of resources can be found [here](#).
- Browse the current astronomy news. Is there something unique happening in the sky where you can measure something interesting? (Again, make sure you have a quantitative science goal.)

Your proposal has to be distinct from your lab-mates, so be sure to check with them. They will automatically be Co-Investigators on your proposal.

Figure out the technical aspects of your project. When is your target observable? What filter or grating do you need? What should the exposure times be? Use your experience from the previous labs, and the manuals of the instruments as guides. Read the lab manual for Lab 2 of the same observation type.

Use the latex template provided below to write your proposal. Note that at most 1 page can be used for the scientific justification. The other pages may be used for figures, tables, and object list. You have to use an 11-pt font, 1-inch margins, and single spacing, otherwise your proposal will be disqualified!

Make sure to submit your proposal by the deadline - late submissions cannot be accepted!

The proposal review will be blinded - for the PI and Cols, list only your SBU IDs, no names!

[LaTeX template file](#)

[LaTeX style file](#)

[Reference file for template](#)

[Figure for template](#)

[PDF of template](#)

- [Lab 1: CCDs](#)
- [Lab 2.1: Exoplanet transit](#)
- [Lab 2.2: Spectroscopy](#)
- [Lab 3: Your own proposal](#)
- [Lab 4: Interferometry](#)
- [Astronomical Data Archives](#)
- [Weather](#)
- [End-of-night report](#)

Computing

- [Computing Resources](#)
- [Astro Software Overview](#)
- [Bash](#)
- [LaTeX](#)
- [Python](#)
- [Jupyter](#)
- [GitHub](#)
- [ds9](#)
- [SExtractor](#)
- [Topcat](#)
- [Astrometry.net](#)
- [Image arithmetic \(+ftools\)](#)
- [Stacking images](#)

Clone this wiki locally

<https://github.com/sibirrer/PH>



→ **Catchy Title**

→ **PI:** put your SBU ID here

Abstract of Scientific Justification (*will be made publicly available for accepted proposals*):

→ Your abstract is the review panel's window into your proposal: the abstract provides an initial impression about your proposal and it is also what panel members refer to at the review meeting to remind themselves about the content of your proposal. Take advantage of the opportunity to give the panel members an understandable and concise summary of what you want to do, and why. Write your abstract so that non-specialists can quickly understand why the observations you want to make are important.

Summary of observing runs requested for this project

Run	Telescope	Instrument	Time Request	Moon	Optimal months	Accept. months
1	14-inch	imaging	3x 2 hours	dark	Oct	Oct - Nov
2	14-inch	spectrograph	1 night	grey	Mar	Mar - Apr
3						
4						
5						
6						

Scheduling constraints and non-usable dates (*up to six lines*).

→ Run 1 needs to be scheduled to catch phase 0 of the eclipsing binary, on Oct. 20, Oct. 25, Oct. 30 or Nov. 4. Run 2: please avoid Apr. 1 (proposers have a mid-term the next morning).

Scientific Justification

Be sure to include overall significance to astronomy. For standard proposals limit text to one page with figures, captions and references on no more than two additional pages.

The scientific justification should explain the overall goals of your program in the context of your field, as well as the importance of your program to astronomy. Writing a good scientific justification is an art. It takes skill and practice. And it requires a good scientific idea. This last you must supply but a few general guidelines about proposal writing might still be helpful...

- State succinctly and clearly the problem you are trying to solve and the progress that will be made toward doing so if the proposed observations are successful. If the review panel members have to work hard even to understand what you want to do, they are unlikely to be sympathetic to your proposal.
- Explain clearly why the project is important and how it relates to the broad context and important issues in your field. Many proposals focus too tightly on a specific observational goal (e.g. “measure the velocity dispersion of this cluster of galaxies”) without explaining why it is important or how it relates to a significant question about the Universe.
- Be specific. If your observations will “constrain theoretical models,” then discuss what will be constrained and why those constraints matter. Make sure the review panel understands exactly why the observations you propose will make a difference in your field, and exactly how the observations will refine or require changes in the theory.
- Keep it simple. Try to focus on the central idea of your proposal. Complex arguments are hard to explain and hard for the panel members to follow. Distracting tangential arguments obscure the theme of your proposal.
- Include a figure to help explain what you want to do. Sample data or model predictions shown in a figure often help clarify complex arguments for the panel members. A sample figure is included below with this proposal.
- Keep it short. Never exceed a page for the text of the scientific justification, and never reduce the font size. It may even help to be a little under a page, and increase the font size a little! Organize your presentation with paragraphs, headings, and bullets so it is easy to read.
- Include and check references as appropriate [Bell et al., 1996].
- Print out the proposal to be sure your LaTeX is correct. Proofread it. Make sure the proposal is correct scientifically, technically, and grammatically. Run a spellchecker.

Finally, when an opportunity arises, volunteer to serve on a TAC or review panel. The experience is a great help in learning how to write a good scientific justification.

References

D. J. Bell, C. D. Biemesderfer, J. Barnes, and P. Massey. An Automated System for Receiving KPNO Proposals by Electronic Mail. In G. H. Jacoby and J. Barnes, editors, *Astronomical Data Analysis Software and Systems V*, volume 101 of *Astronomical Society of the Pacific Conference Series*, page 451, 1996.

Scientific Justification

1-page limit!

Scientific Justification

- describe your project to a knowledgeable, but non-expert audience
- provide background information / motivation
- what is the goal of your measurement?
- make it exciting / important!
- polish the text - typos and carelessness are distracting

2 pages for figures,
references, etc.

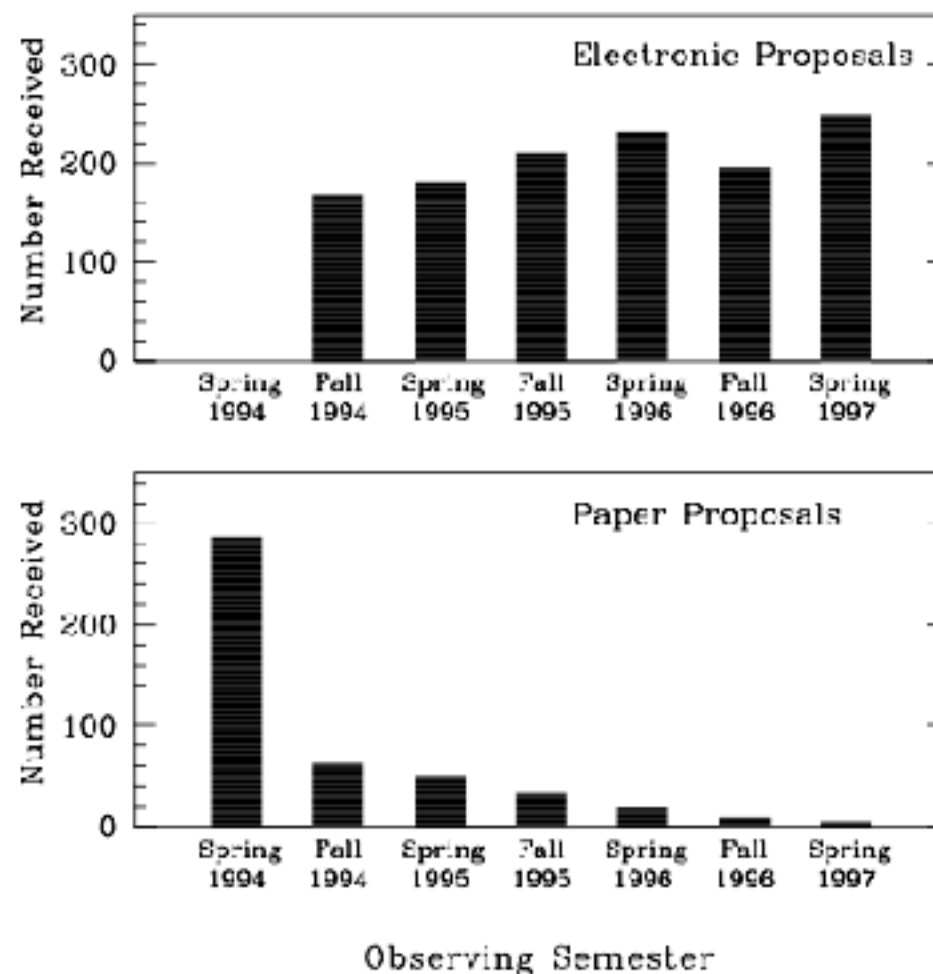


Figure 1: This sample figure shows how quickly electronic proposals for telescope time replaced paper ones.

Experimental Design / Technical Justification

Mt. Stony Brook Proposal

Page 4

This box blank.

→ **Experimental Design** Describe your overall observational program. How will these observations contribute toward the accomplishment of the goals outlined in the science justification? If you've requested long-term status, justify why this is necessary for successful completion of the science. (limit text to one page)

The review panel looks to this section to find out about the overall strategy of your data analysis. You need to convince the panel that your strategy will be successful. If the strategy is found to be infeasible, the proposal will be outright rejected.

- What are the target objects and target datasets, and how were they selected?
- Comment on the observability of your targets.
- Provide a **detailed explanation of the exposure time estimate**. Where possible, scale from your existing data to calculate the required exposure time. Also provide an estimate of how long your total observing session will take (with multiple exposures, calibration, etc.).
- Provide an explanation of which filter(s) you will use.
- What calibration data will you require?
- Will you require external data? If so, describe the nature of that dataset, and how you will analyze it.
- What do you plan to measure, and how?

1-page limit

Technical Justification

- as important as your scientific justification is the technical justification
- if your project is not feasible, it will be rejected
- from the exoplanet lab, you have an idea of what magnitude star requires what exposure time
 - can scale to other magnitudes with CCD signal-to-noise equation (recall that $mag \propto \log[flux]$)
 - make sure to justify choice of filter(s), too

to overlay external catalogs:

The image displays two software windows. The left window, titled 'USNO-B1.0', shows a catalog table with columns for object ID, B magnitude, R magnitude, I magnitude, and other parameters. A red arrow points to the 'Filter' field, which contains the expression '\$R1mag>0 && \$R1mag<14'. Below the table, there are buttons for 'Retrieve', 'Cancel', 'Filter', 'Clear', 'SAMP', and 'Plot'. The right window, titled 'SAOImage ds9', shows a large astronomical image with numerous green circles overlaid, representing cataloged objects. A red arrow points to the 'Catalogs' menu item in the 'Analysis' menu, which is open, showing a list of available catalogs including USNO-B1.0.

USNO-B1.0

File Edit Catalog Server Name Server Symbol Preferences

Catalog

Name USNO-B1.0

Identification 1/284/001

Reference 001

IAU Location Code 500

Object

Name

RA 20:00:39.091 S 122:40:46.202 1k5 Update

Width 28.88216 Height 28.88591 Admin

Table

Filter \$R1mag>0 && \$R1mag<14 Edit

Sort Increase Decrease

Max Rows 5000 Found 696

RAJ2000 DEJ2000

ID	B1mag	R1mag	I1mag	R2mag	I2mag
0	15.41	10.88	12.43	10.79	9.97
1	16.00	11.24	15.09	13.18	12.76
2	14.87	12.95	14.26	13.08	11.11
3	15.41	13.97	14.04	13.00	13.64
4	16.60	13.33	15.32	12.99	12.34
5	15.33	13.68	14.79	13.53	12.83
6	11.54	10.71	11.28	10.68	10.43
7	16.66	13.87	15.76	14.01	13.11
8	12.54	11.91	12.35	11.89	11.70
9	13.34	10.45	12.12	10.30	9.56
10	17.19	13.94	15.48	13.18	13.52
11	16.47	11.24	15.57	13.46	11.49
12	12.58	9.66	11.33	9.59	8.77
13	14.20	12.36	13.30	11.32	11.40
14	16.30	13.72	15.32	13.27	13.10
15	15.96	13.60	14.88	12.98	12.77
16	14.93	11.01	15.21	13.11	12.29
17	14.66	13.72	19.26		
18	12.60	10.10	16.37	12.33	
19	14.20	11.14	13.52	11.01	10.44
20	13.42	12.10	12.53	11.19	11.30
21	15.90	13.51	14.88	13.03	12.53
22	13.68	12.42	12.73	11.87	11.47

Status Row 25

Retrieve Cancel Filter Clear SAMP Plot

SAOImage ds9

File Edit View Frame File Zoom Scale Color Region WCS Analysis Help

File science 3.DFA.fits

Object

Value

WCS

Physical X Y

Image X Y

Frame 1 x 2 0

File edit view frame

zoom in zoom out zoom fit

Block Block Parameters...

Smooth Smooth Parameters...

Crosshair Parameters...

Name Resolution...

Image Servers Archives

Catalogs

Catalog Tool...

Line Plot Tool...

Bar Plot Tool...

Scatter Plot Tool...

Virtual Observatory...

Web Browser...

Analysis Command Log

Load Analysis Commands...

Clear Analysis Commands

Search for Catalogs

Clear All

Match

Database

Optical

Infrared

High Energy

Radio

Observation Logs

ASCC-2.5

AMVO

Carlsberg Meridian 14

CSC 1.2

CSC 2.2

CSC 2.3

AC 2000.2

NOMAD

PMX

SAO J2000

SDSS Release 5

SDSS Release 6

SDSS Release 7

SDSS Release 8

SDSS Release 9

Tycho 2

USNO-A2.0

USNO-B1.0

USNO UCAC3

might need to filter results

Analysis Plan

Analysis Plan *Describe the steps that you will take to analyze the data. State which tasks will be due for the weekly check-ins (weeks 1, 2, 3). The week 4 task will be to complete the lab report. In case you have less than four weeks between taking the observations and the last day to submit lab reports, how will the analysis plan change?*

+1 week:

+2 weeks:

+3 weeks:

+4 weeks: Submit the lab report.

note: last day to submit lab reports: Dec 7

Observing Run Details for Run 1: 14-inch/STL-1001E

Technical Description

Describe the observations to be made during this observing run. Justify specific instrument choices such as the required filter, or the grating choice, as well as the requested lunar phase. List targets, coordinates, and magnitudes (or surface brightness, if appropriate) in the Target Tables section below.

Create a separate "run" page for each telescope/instrument combination being used for your project.

Instrument Configuration

Filters: B,V,R,I,H α
 Grating/grism: -
 Order: -
 Cross disperser: -

Slit: -
 Multislit: -
 λ_{start} : -
 λ_{end} : -

Fiber cable: -
 Corrector: -
 Collimator: -
 Atmos. disp. corr.: -

R.A. range of principal targets (hours): 12 to 13

Dec. range of principal targets (degrees): 10 to 15

Special Instrument Requirements

Describe briefly any special or non-standard usage of instrumentation.

Target Table for Run 1: DECcam

Obj ID	Object	α	δ	Epoch	Mag.	Filter	Exp. time	# of exp.	Lunar days	Sky Seeing	Comment
777	NGC 7078	21:30:10	12:10:03	2000.0	18.6	g	1000	5	4		globular cluster
778	NGC 7078	21:30:10	12:10:03	2000.0	18.6	r	1000	5	4		same cluster
779	NGC 7078	21:30:10	12:10:03	2000.0	18.6	i	1000	5	4		

AST 443 / PHY 517 proposals

proposal template available on github:

https://github.com/sibirrer/PHY517_AST443/wiki/Your-own-telescope-proposal

blind review: list only your SBU ID as author!

github page includes links to example telescope proposals

Time Allocation Committee

Time Allocation Committees

- proposals are reviewed by panels of researchers, chosen by the responsible agencies (e.g. HST - STScI)
- panels are assembled by topical groups (e.g. cosmology)
- every panelist has to read every proposal assigned to that panel, typically ~80
- preliminary grades submitted online
- 2-day meetings to discuss all proposals and finalize grades

AST 443 / PHY 517 TAC

- you will be assigned a list of ~6 proposals to evaluate and grade
- you will have to send in grades and comments for all proposals on your list a few days before the TAC meeting

AST 443 / PHY 517 TAC

- you will be primary reviewer for one proposal, and secondary reviewer for another proposal
- during the TAC meeting, the primary and secondary reviewers will lead to discussion of each proposal, but *everybody will be expected to take part*
- the PI of the proposal and their collaborators, as well as PIs of directly competing proposals, will leave the room

AST 443 / PHY 517 TAC

- after each discussion, you will re-grade the proposal via secret ballot
- we will rank the proposals based on the final grade
- *each group will work on its top-ranked project*
- after the TAC meeting, the primary and secondary reviewer will collate the comments into a final evaluation of that proposal

**Blind reviews:
avoiding unconscious bias**

Avoiding unconscious bias

- Unconscious / implicit bias: our judgment is biased by stereotyped expectations
- has been well documented in much of society

Image: Texas Medical Center



Avoiding unconscious bias

- Unconscious bias is “unconscious” - there is no malicious intent (unlike racism / sexism etc.)
- Result of how humans make decisions → difficult (but not impossible!) to avoid



Affinity Bias

Feeling a connection to those similar to us



Perception Bias

Stereotypes and assumptions about different groups



Halo Effect

Projecting positive qualities onto people without actually knowing them



Confirmation Bias

Looking to confirm our own opinions and pre-existing ideas.

Avoiding unconscious bias

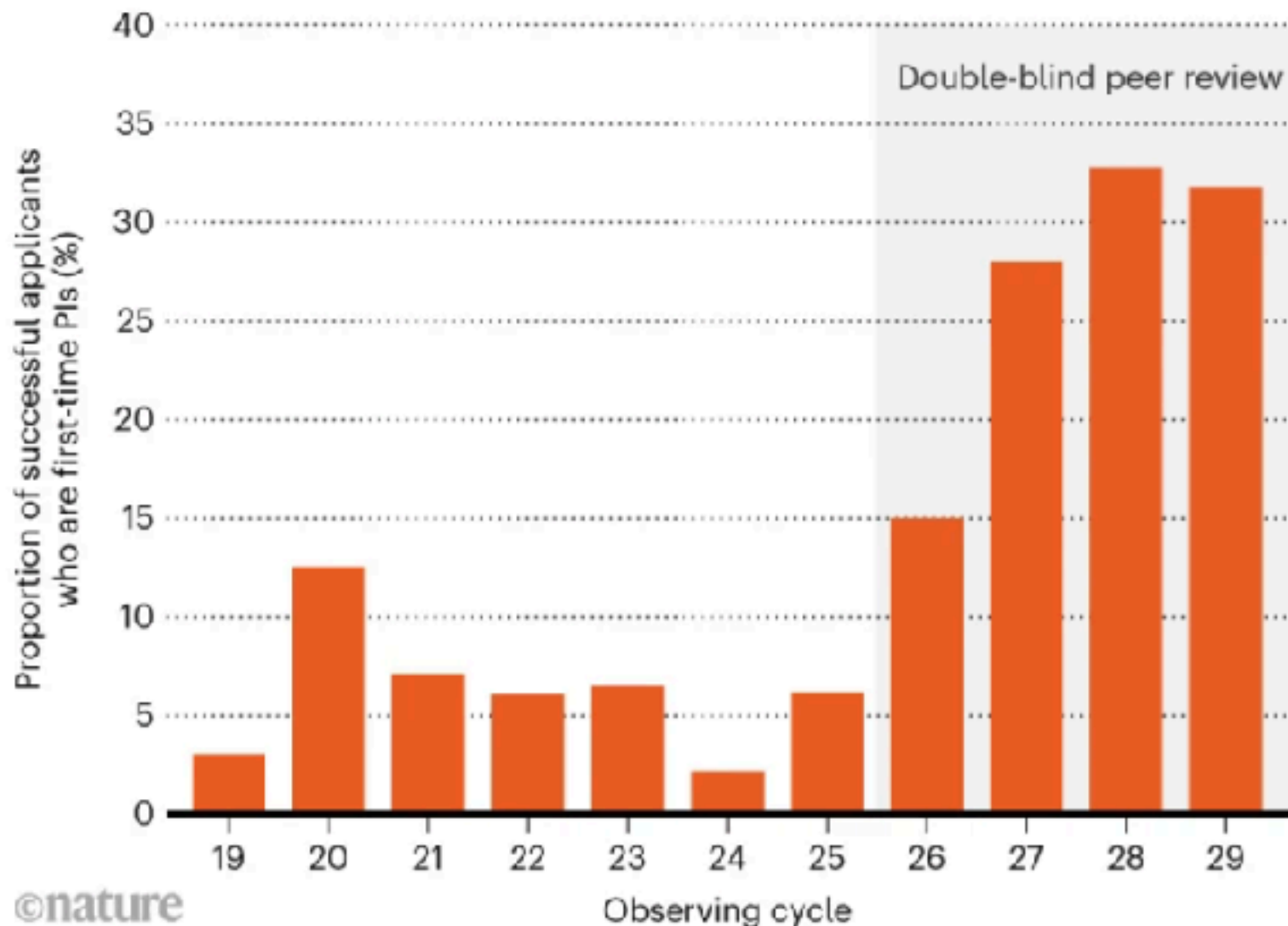
- In 2018, the Hubble TAC was conducted dual-anonymously
- PIs were not identified; had to avoid identifying themselves in the narrative
- Reviewers were told not to guess the proposers

Avoiding unconscious bias

- In dual-anonymous review: seniority bias could be overcome

FIRST-TIME OBSERVERS

Since the introduction of double-blind peer review for proposals in 2018 (cycle 26), higher numbers of new principal investigators have won observing time on the Hubble Space Telescope.



Nature news,
2021

- Implemented also for several other review processes

Therefore:

- class TAC: double-blind
- class grading: as blind as we can make it
- “we” are still trying to figure this out
- be conscious of biases when you get into advisorship / leadership roles

Further Reading

- Reid 2014: Gender-based Systematics in HST Proposal Selection. <https://ui.adsabs.harvard.edu/abs/2014PASP..126..923R/abstract>
- Strolger & Natarajan 2019: Doling out Hubble time with dual-anonymous evaluation. <https://physicstoday.scitation.org/doi/10.1063/PT.6.3.20190301a/full/>
- Caplar, Tacchella, Birrer 2017: Quantitative evaluation of gender bias in astronomical publications from citation counts. <https://ui.adsabs.harvard.edu/abs/2017NatAs...1E..141C/abstract>